Indian Institute of Technology Dharwad

Department of Computer Science & Enginnering

Lab Assignment 5

Event Simulator for ToyRICS

CS301 ---- Computer Architecture

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Benchmark Results

Program	*Instructions*	*Cycles*	*IPC*	*CPI*
Even Odd	*6*	*249*	*0.024096*	*41.5*
Palindrome	*49*	*1979*	*0.024760*	*40.38*
Prime	*29*	*1173*	*0.024723*	*40.44*
Fibonacci	*78*	*3174*	*0.024575*	*40.69*
Descending Sort	*277*	*11469*	*0.024152*	*41.40*

Observation & Conclusion :

The introduction of discrete event model helps us to model our processor to replicate a real-life processor even more, owing to the fact that for any real life scenario, the computations like ALU operations (mult, div, etc.) or memory access would not be infinitely fast, they would have a finite time lag which is significant especially when compared to the time for a cycle and hence introducing the same helps to predict and model the functionality of a real life processor, better!

- *Main Memory Latency has been modeled. It is reflected in IF stage and during the load/ stores processes.*
- *Latencies of different functional units have been modeled. This is reflected in EX stage, and includes components such as ALU, multiplier, divider, etc.*
- *Main memory latency is observed to be roughly around 40% for our processor configuration. (The general latency could be anywhere between 20-50%, and hence our observations are in line with the general trends.*
- *If programs have more data hazards and control hazards, then it will take more cycles per instruction*